Methods of finding correlation Coefficient

1. Karl person's product movement method.
2. Speckers man's rank correlation Coeff.
3. Scatter diagram method.
4. Coeff of concurrent deviations.

Karl person's method.

One that is most common is the Pearson correlation coefficient, which is represented as r.

$$r = \frac{Cov\ (x, y)}{\sigma_x \sigma_y}$$

$$cov\ (x, y) = \frac{\sum (x - \bar{x}\ )(x - \bar{y})}{n}$$

$$\sigma_x = \sqrt{\frac{\sum(x-\overline{x})^2}{n}} \qquad \sigma_y = \sqrt{\frac{\sum(y-\overline{y})^2}{n}}$$

FORMULA 2

$$r = \frac{n\sum xy - \sum x \sum y}{\sqrt{n\sum x^2 - \left(\sum x\right)^2}\ \sqrt{n\sum y^2 - \left(\sum y\right)^2}}$$

Correlation coefficient range

- The values of a **Pearson correlation coefficient** range between -1 and 1.
- A correlation of $r = -1$ shows a **perfect negative** correlation.
- A correlation of $r = 1$ shows a **perfect positive** correlation
- A correlation of $r = 0$ shows **no linear relationship** the two variables.

Correlation Coefficient (r)	Description (Rough Guideline)
+1.0	Perfect positive + association
+0.8 to 1.0	Very strong + association
+0.6 to 0.8	Strong + association
+0.4 to 0.6	Moderate + association
+0.2 to 0.4	Weak + association
0.0 to +0.2	Very weak + or no association
0.0 to -0.2	Very weak- or no association
-0.2 to – 0.4	Weak- association
-0.4 to- 0.6	Moderate - association
-0.6 to - 0.8	Strong- association
-0.8 to-1.0	Very strong -association
- 1.0	Perfect negative association

Correlation

Question: Find coefficient of correlation of the following data by direct method

x	y
9	15
8	16
7	14
6	13
5	11
4	12
3	10
2	8
1	9

$$r = \frac{n\sum xy - \sum x \sum y}{\sqrt{n\sum x^2 - (\sum x)^2}\ \sqrt{n\sum y^2 - (\sum y)^2}}$$

$$n = \text{number of observation} = 9$$

$$\sum xy = ?$$

$$\sum x = ?$$

$$\sum y =$$

$$\sum x^2 =$$

$$\sum y^2 =$$

Coefficient of correlation by direct method

$\sum xy$

$\sum x$

$\sum y$

$\sum x^2$

$\sum y^2$

x	y	x^2	y^2	xy
9	15	81	225	135
8	16	64	256	128
7	14	49	196	98
6	13	36	169	78
5	11	25	121	55
4	12	16	144	48
3	10	9	100	30
2	8	4	64	16
1	9	1	81	9
$\sum x = 45$	$\sum y = 108$	$\sum x^2 = 285$	$\sum y^2 = 1356$	$\sum xy = 597$

Coefficient of Correlation

$$r = \frac{n\sum xy - \sum x \sum y}{\sqrt{n\sum x^2 - (\sum x)^2}\ \sqrt{n\sum y^2 - (\sum y)^2}}$$

$\text{n} = no.of\ pair\ of\ observers = 9$

$$\sum xy = 597$$

$$\sum x = 45$$

$$\sum y = 108$$

$$\sum x^2 = 285$$

$$\sum y^2 = 1356$$

$$r = \frac{9(597)-(45)(108)}{\sqrt{9\,(285)-(45)^2}\ \sqrt{9(1356)-(108)^2}}$$

$$r = \frac{5373 - 4860}{\sqrt{540}\ \sqrt{540}} = \frac{513}{440} = 0.95$$

It is strong positive correlation

Coefficient of Correlation by Shortcut method

x	y
9	15
8	16
7	14
6	13
5	11
4	12
3	10
2	8
1	9

$$r = \frac{n\sum xy - \sum x \sum y}{\sqrt{n\sum x^2 - (\sum x)^2}\ \sqrt{n\sum y^2 - (\sum y)^2}}$$

$$r = \frac{n\sum dxdy - \sum dx \sum dy}{\sqrt{n\sum dx^2 - (\sum dx)^2}\ \sqrt{n\sum dy^2 - (\sum dy)^2}}$$

$$dx = deviated\ x = x - A$$

Where A is assumed mean of x

$$dy = deviated\ y = y - B$$

Where B is assumed mean of Y

$n = ?$

$\sum dxdy = ?$

$\sum dx = ?$

$\sum dy^2 = ?$

$\sum dx^2 = ?$

$\sum y^2 = ?$

coefficient of correlation by shortcut method

$dx = x - 5$ $dx = y - 11$

x	y	dx	dy	dx^2	dy^2	$d$xdy
9	15	4	4	16	16	16
8	16	3	5	9	25	15
7	14	2	3	4	9	6
6	13	1	2	1	4	2
5	11	0	0	0	0	0
4	12	-1	1	1	1	-1
3	10	-2	-1	4	1	2
2	8	-3	-3	9	9	9
1	9	-4	-2	16	4	8
		$\sum dx = 0$	$\sum dy = 9$	$\sum dx^2 = 60$	$\sum dy^2 = 69$	$\sum dxdy = 57$

Correlation Coefficient

n = $no. of\ pair\ of\ observers = 9$

$$r = \frac{n\sum dxdy - \sum dx \sum dy}{\sqrt{n\sum dx^2 - (\sum dx)^2}\ \sqrt{n\sum dy^2 - (\sum dy)^2}}$$

$$\sum dxdy = 57$$

$$\sum dx = 0$$

$$\sum dy = 9$$

$$\sum dx^2 = 60$$

$$\sum dy^2 = 69$$

$$r = \frac{9(57)-(0)(9)}{\sqrt{9\,(60)-(0)^2}\ \sqrt{9(69)-(9)^2}}$$

$$r = \frac{513}{\sqrt{540}\ \sqrt{540}} = \frac{513}{540} = 0.95$$

Strong positive Correlation